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Academic Year: 2025-2026

Department of Artificial Intelligence and Machine Learning

Date: 17/12/2025	Test – II	Max. Marks : 10 + 50
Semester: 5th	UG	Duration : 2 hours
Course Title: Mathematical Algorithm for AI		Course Code: AI255TBC

Part – A

QN	Questions	M	BT	CO
1.1	If for two vectors u, v, u.v = 0, then (a) u and v are collinear (b) u and v are orthogonal (c) u and v are linearly dependent (d) u and v have no relation	2	2	2
1.2	If the characteristic equation of a 3x3 matrix A is $\lambda(\lambda^2 - 4) = 0$ then the determinant of A is (a)0 (b) 4 (c) - 4 (d) indeterminate.	2	3	3
1.3	The orthogonal complement subspace of $\mathbb{R}^2$ is (a) the z-axis (b) the cross product of the basis vectors for $\mathbb{R}^2$ (b) (c) $\left\{\begin{pmatrix}0\\0\end{pmatrix}\right\}$ (d) indeterminate	2	3	2
1.4	Which of the following is not a property of dot product? Here, a, b, c are vectors and k a real scalar. (a) a.b = b.a (b) a.(b+c) = a.b + a.c (c) a.(kb + c) = k(a.b) + a.c d) a.(b.c) = (a.b).c	2	2	3
1.5	If u, v, and w are orthonormal vectors, then $3u^Tv - 2v^Tv + 3u^Tw + 3w^Tw$ is equal to (a) 1 (b) 7 (c) 11 (d) a value that cannot be computed without knowing the length of the vectors.	2	3	2

Part – B

SL. No	Questions	M	BT	CO
1.	Let u, v, w be three vectors in $\mathbb{R}^4$. You are told that $u \cdot v = u \cdot w$ Assert or Reject the following statement with proper justification: Since $u \cdot v = u \cdot w$, the vector $v = w$ vector.	4	2	1
2.	Find the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix}5 & 4\\4 & 5\end{pmatrix}$	6		1

3. Let W be the vector space spanned by the columns of A, where A is as given below. Find the Orthogonal Complement subspace of W.

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix}$$

6 | 3 | 1

4. Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Find the null space and column space of A. Specify the dimension of the null space of A and column space of A. What do you observe? Can you generalize this result for any square matrix?

10 | 3 | 2

5. For the matrix A given below, obtain the four fundamental subspaces associated with it.

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix}$$

You are expected to identify a suitable basis for each of the subspaces and the dimension for each of those subspaces. Tabulate your answer as follows:

12 | 2 | 1

Subspace	Basis	Dimension
Null space of A		
Column Space of A		
Null Space of A^T		
Column space of AT		

6. Given the following vectors, check if they are orthonormal or not.

$$\phi_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \phi_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \phi_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \phi_4 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

Express the vector

$$u = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$

as a linear combination of $\phi_1, \phi_2, \phi_3, \phi_4$.

12 | 3 | 2